



Full length article

Enhancing stock return prediction in the Chinese market: A GAN-based approach

Hongxu Wu^a, Qiao Wang^{b,c,*}, Jianping Li^a, Zhibin Deng^{a,d}

^a School of Economics and Management, University of Chinese Academy of Sciences, Beijing, China

^b Institute of Science City Industry, University of Chinese Academy of Sciences, Beijing, China

^c College of Resources and Environment, University of Chinese Academy of Sciences, Beijing, China

^d MOE Social Science Laboratory of Digital Economic Forecast and Policy Simulation at UCAS, China

ARTICLE INFO

Keywords:

Stock return prediction
Chinese market
Factor investment
Generative adversarial network
Machine learning

ABSTRACT

With the rapid advancement of machine learning technologies, the accuracy of predictive models has seen continuous improvement. In this paper, we aim to apply machine learning models to predict stock market returns and implement factor investment in the Chinese A-share market. We propose a novel GAN model integrating LSTM, attention mechanisms, and CNN to address time-dependence issues and low signal-to-noise ratios more effectively in stock return prediction. Out-of-sample results demonstrate that the proposed GAN model outperforms other benchmark models in terms of predictive performance and exhibits superior generalization capabilities. Furthermore, our analysis of predictor importance highlights the pivotal role of transaction and profitability metrics in predicting returns in the Chinese A-share market. Additionally, when incorporating macroeconomic predictors into the training data, our model demonstrates stability amidst macroeconomic fluctuations. These empirical findings underscore the potential value of the proposed GAN model in the Chinese stock market investment.

1. Introduction

China's A-share stock market has become the world's second-largest stock market after the US, with a total market capitalization of \$14.6 trillion USD (CNY 87.8 trillion) as of December 2022. The Chinese stock market has received great attention from both domestic and foreign investors and researchers. In contrast, its unique characteristics, such as retail investors' domination, high sensitivity to regulatory policies, and immaturity, indicate that there are profound differences with the advanced stock markets, and stock return predictors and trading strategies that work in the advanced markets may lose efficiency in this market. Meanwhile, with the explosion of research on machine learning techniques and their applications to finance, the growing literature in this field predominantly focuses on the US stock market (Kozak et al., 2020; Gu et al., 2020), and it is still unclear whether those machine learning methods, predictors and strategies are efficient in China's stock market. Based on this background, this paper contributes to China's empirical asset pricing literature by employing machine learning models to predict stock market returns and make factor investment in the Chinese stock market.

It has always been an active area of research in finance to predict stock returns. Traditional research approaches focus on factor selection and uncovering of anomalies. Since (Fama and French, 1993) published the first three-factor model, various multi-factor models (Carhart, 1997; Novy-Marx, 2013; Fama and French, 2015; Stambaugh and Yuan, 2017; Daniel et al., 2020) have been proposed to explain stock returns in the past thirty years. These endeavors explored new factors from different perspectives such

* Corresponding author at: Institute of Science City Industry, University of Chinese Academy of Sciences, Beijing, China.

E-mail address: wangqiao@ucas.edu.cn (Q. Wang).

<https://doi.org/10.1016/j.ribaf.2025.102760>

Received 5 May 2024; Received in revised form 8 October 2024; Accepted 6 January 2025

Available online 12 January 2025

0275-5319/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

as financial markets, corporate finance economics and behavioral finance. The common practice in the traditional approaches is to select a small number of factors (usually three to five) to construct the linear multi-factor models.

Although these “ad-hoc” sparse models help to explain the underlying economic and financial theories well, there are doubts regarding their accuracy in this era of big data. As financial economists have already identified a great many predictors for the cross-section of stock returns (Green et al., 2013; Hou et al., 2015; Kozak et al., 2020), traditional econometric-linear multi-factor models are naturally deficient in terms of accuracy due to the small number of factors considered. Furthermore, since the high-dimensional nature of data introduces complex interactions and nonlinear structures that cannot be adequately captured by traditional linear models, inappropriately employing linear models can create substantial errors in return predictability (Pohl et al., 2018).

To overcome these limitations associated with traditional linear multi-factor models, a growing body of literature has introduced machine learning methods into empirical financial analysis to efficiently capture high dimensional, non-linear patterns and regularities of data. These attempts offer researchers the opportunity to consider more flexible functional form specifications and models, which could probably yield promising results in return prediction. Machine learning methods are generally divided into three categories: supervised learning, unsupervised learning and reinforcement learning. Regarding their applications in empirical research related to the stock market, these three machine learning methods specialize in solving different issues. Unsupervised learning methods are generally used to reduce the dimensionality of firm characteristics and other predictors (Kelly et al., 2019; Lettau and Pelger, 2020; Bybee et al., 2022; Ma et al., 2023), and reinforcement learning methods are usually used to guide trading decisions (Pricope, 2021). Supervised learning methods, which aim to obtain the correct output from a given labeled training data, are suitable for return prediction, which is the target of this paper. For example, Gu et al. (2021) and Feng et al. (2024) employ supervised learning methods (including the conditional auto-encoder model based on conditional beta pricing model, and feedforward neural network) for stock return prediction, and show encouraging results with out-of-sample pricing errors much smaller than other factor models, such as Fama–French multi-factor models, principal component analysis (PCA) and instrumented principal component analysis (IPCA).

In recent years, many supervised learning models have been developed to predict stock returns. In particular, the generative adversarial network (GAN)—an advanced neural network, since proposed by Goodfellow et al. (2014), has been studied and employed extensively in the field of financial economics to predict stock returns. For example, Zhang et al. (2019) train a GAN model to predict the prices of various stock indices and individual stocks, and their results show that GAN outperforms other machine learning models such as long short-term memory (LSTM) and support vector machine (SVM) in terms of mean absolute error (MAE) and other metrics. Kumar et al. (2022) propose a phase-space reconstruction (PSR) method to reconstruct price series and combine it with GAN for prediction, and their version of the GAN model achieves better prediction accuracy and shorter training time than LSTM. Chen et al. (2024) introduce feedforward neural network (FFN), recurrent neural network (RNN) and LSTM into the GAN model for the estimation of stochastic discount factor (SDF) parameters, and the results show that this ingenious GAN model outperforms FNN in terms of out-of-sample Sharpe ratios, explained variation and pricing errors. The existing literature shows the potential of GAN models to predict stock returns, and highlights their superiority over traditional models and other supervised learning methods in terms of accuracy and performance metrics.

While the results of GAN models are encouraging, research on model improvement and their suitability for finance has never been stopped. Our paper contributes to this rapidly-emerging literature by proposing a new version of GAN model to improve its suitability for financial data. Specifically, we include three new components in the proposed GAN model. First, we incorporate LTSM in the generator network to address the temporal dependence of stock returns and to mitigate common issues of neural networks, such as gradient explosion or vanishing. In addition, given the low signal-to-noise ratio of financial data, we introduce the self-attention mechanism in the generator network to extract critical information and reduce the risk of in-sample overfitting and out-of-sample error explosion. Meanwhile, in order to improve the processing efficiency of high-dimensional features, we use a convolutional neural network (CNN) as the discriminator to reduce features. As in related literature (Gu et al., 2020; Li and Mei, 2020; Leippold et al., 2022; Chen et al., 2024), we also analyze various models and assess the performance of our proposed GAN model by comparing it with five other benchmark models (OLS, ENet, NN3, LSTM, and a specific GAN used based on Chen et al. (2024)) in terms of return prediction accuracy and Sharpe ratios of portfolios. The empirical results show that our new GAN model outperforms all the benchmarks, including the two latest models: the 3-layer neural networks (NN3) in Gu et al. (2020) and the GAN model in Chen et al. (2024).

The GAN models, although a more advanced and better-performing method of machine learning, still face limitations in terms of interpretability. The second main contribution of this paper attempts to address this shortcoming. Firstly, we investigate the importance of different types of predictors (including firm, financial market and macroeconomic predictors) for stock return prediction, to gain insights into the underlying factors and mechanisms driving stock market dynamics. Our empirical results show that trading-related and profitability-related predictors exhibit prominent performance. The high importance of transaction-related predictors is consistent with the longstanding state of Chinese stock market which is not yet fully efficient and experiences market frictions. The high contribution of profitability-related predictors, which is usually consistent with value investing, indirectly indicates important influences of foreign institutional investors (the so-called northbound capital and wind vane for China's A-Share stock market) for China's stock market. Secondly, we construct long-short and long-only portfolios based on the model's predictions to evaluate its economic performance. This provides useful guidance for investment strategies in the Chinese A-share market and demonstrates the potential practical value of the proposed GAN model. Our results show that these portfolios generated by the proposed GAN model outperforms the benchmarks in terms of Sharpe ratios.

The third contribution of this paper is that we investigate the impact of different macroeconomic states on the performance of the proposed GAN model and re-test the performance of the model trained with the inclusion of macroeconomic predictors. In this

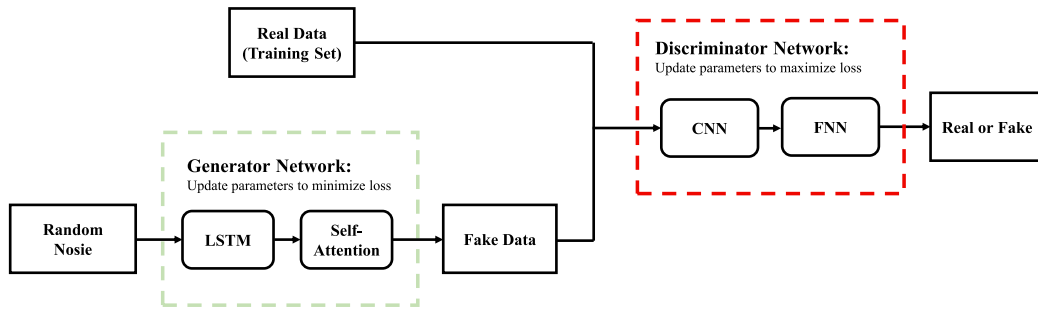


Fig. 1. Structure of the proposed GAN model.

way, the re-trained GAN model will capture a wider range of potential predictors that may affect stock returns, thus enhancing the robustness and applicability of the model in real-world investment scenarios. Our results show that the inclusion of macroeconomic predictors indeed further improves the predictive performance and reduces the model's sensitivity to macroeconomic fluctuations.

The remaining sections of this paper are organized as follows. Section 2 presents the proposed GAN model. Section 3 describes the classification of data and predictors used in the empirical analysis. Section 4 presents the empirical results and the economic insights from the results, and Section 5 concludes.

2. Methodology

In this paper, we assume that the stock return follows the general additive error model:

$$r_{i,t+1} = \mathbb{E}_t(r_{i,t+1}) + \varepsilon_{i,t+1}, \quad (1)$$

where

$$\mathbb{E}_t(r_{i,t+1}) = g(z_{i,t}), \quad (2)$$

$i = \{1, \dots, N\}$ is the index for individual stock, $t = \{1, \dots, T\}$ is the index for monthly time period, $r_{i,t+1}$ is the realized return for stock i at month $t + 1$, $\mathbb{E}_t(r_{i,t+1})$ is the expectation of $r_{i,t+1}$, $\varepsilon_{i,t+1}$ is the random error, $z_{i,t}$ is a p -dimensional vector of predictors, and $g(\cdot)$ is a smooth function, which is unknown but could take the nonlinear structure and interaction terms of predictors into account. Due to the high nonlinearity and complexity of the function $g(\cdot)$, we propose a GAN model to approximate $g(\cdot)$ so that the model could have enough explanatory strength for out-of-sample realized $r_{i,t+1}$.

Specifically, we design a new GAN model, whose structure is shown in Fig. 1. The detailed description for each component of the proposed model is given in the following subsections.

2.1. Generative adversarial network (GAN)

A basic GAN consists of two modules: the generator network and the discriminator network. The generator's task is to generate data that resemble real data, while the discriminator is trained to differentiate between the real data and the fake data generated by the generator. The loss function for a GAN model takes the following form:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_x} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))] \quad (3)$$

where D represents the discriminator network, G represents the generator network, x represents real data, p_x represents the distribution of real data, z represents the noise vector sampled from a prior distribution p_z , and $G(z)$ represents the fake data generated by the generator from the noise vector z . The loss function aims to maximize the discriminative ability of D to distinguish real and fake data while simultaneously maximizing the generative ability of G . The training procedure of a GAN model is presented in Algorithm 1.

Through the adversarial training between the generator and the discriminator network in Algorithm 1, the generator network gradually improves its generation capability, making it difficult for the discriminator to accurately distinguish real data from fake data. Through this adversarial process, the generator network can learn the distribution of the real data, and then we obtain a generator for stock return prediction at the end of training.

2.2. Generator network

The generator is crucial for predicting returns and requires the selection of an appropriate machine learning model based on the research problem. Previous endeavors typically used an LSTM as the generator in GAN (Zhang et al., 2019; Kumar et al., 2022; Chen et al., 2024), because stock returns are time series data, and LSTM (an improved model of RNN) can capture the long-range dependence of time series. Fig. 2 shows the structure of a typical LSTM, and Eq. (4) gives the specific mathematical operation.

Algorithm 1 Training of GAN Model.

-
- 1: **Initialization:** Initialize G and D with parameters θ_g and θ_d .
 - 2: **for** number of training **do**
 - 3: **for** number of discriminator network training **do**
 - 4: • Generate n samples $\{z^{(1)}, \dots, z^{(n)}\}$ from the noise distribution p_z .
 - 5: • Take n samples $\{x^{(1)}, \dots, x^{(n)}\}$ from real data x .
 - 6: • Update discriminator network D with stochastic gradient descent:
 - 7: $\nabla_{\theta_d} \frac{1}{n} \sum_{i=1}^n - [\log D(x^{(i)}) + \log (1 - D(G(z^{(i)})))]$.
 - 8: **end for**
 - 9: • Fix discriminator network D and update generator network G with stochastic gradient descent:
 - 10: $\nabla_{\theta_g} \frac{1}{n} \sum_{i=1}^n \log (1 - D(G(z^{(i)})))$.
 - 11: **end for**
-

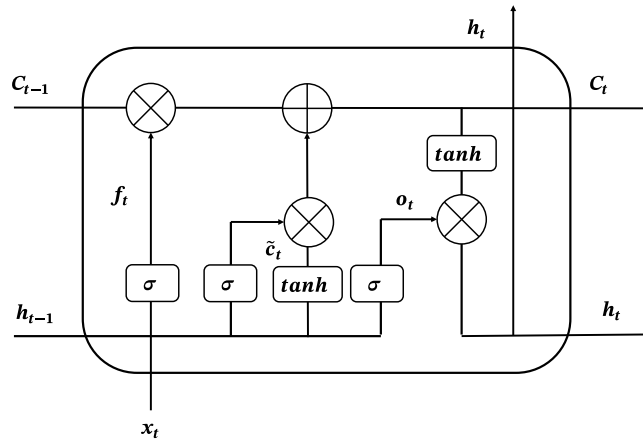


Fig. 2. The structure of an LSTM.

$$\begin{aligned}
 i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\
 f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) \\
 \tilde{c}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c) \\
 o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) \\
 c_t &= f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t \\
 h_t &= o_t \cdot \tanh(c_t)
 \end{aligned} \tag{4}$$

The input gate i_t controls the amount of information to be inputted at the current time step, the forget gate f_t , $\tilde{c}_t$ determines the extent to which past states are retained, and the output gate o_t regulates the amount of information to be outputted at the current time step. c_t and h_t represent long-term memory and short-term memory respectively. By utilizing LSTM as the generator model, we can fully exploit its sequential memory and gating mechanism to predict stock returns effectively.

However, since our inputs are a large number of predictors and the importance of each predictor to the predictive accuracy is bound to be inconsistent, we add an additional self-attention mechanism after LSTM to ensure the out-of-sample performance of the model.

Self-attention is a commonly used technique in natural language processing to selectively focus on different parts of input sequences when processing sequential data. It dynamically assigns importance weights to each input position, allowing the model to focus on key information, especially when dealing with long sequences. Essentially, self-attention is a tool for weighted summation and feature reduction. The expression of self-attention can be represented by Eq. (5):

$$Attention(Q, K, V) = softmax(\frac{Q \cdot K^T}{\sqrt{d_k}}), \tag{5}$$

where, Q , K , and V are matrices obtained by multiplying the input tensor with three weight-sharing matrices, W_Q , W_K , and W_V , respectively. Q represents the query matrix, K represents the key matrix, V represents the value matrix, and d_k is a matrix used to adjust the dimensionality of the dot product. By performing the operation $Q \cdot K^T$, we obtain the attention scores, which reflect the relevance between vectors. Applying the Softmax activation function to normalize these attention scores, we can determine which time steps and features are more important. Finally, we perform element-wise multiplication between the Softmax values and the V matrix to obtain the attention matrix. This operation allows the model to focus more on important information, improving efficiency and avoiding overfitting.

Meanwhile, self-attention is often used in conjunction with recurrent layers such as RNN, LSTM and GRU. Placing self-attention after the recurrent layer allows the model's final decision to be dependent on the feature dimensions that contribute most to the ultimate goal, effectively improving the model's generalization ability and preventing overfitting issues. Therefore, we include a self-attention layer after 2-layer LSTM in the generator.

2.3. Discriminator network

The task of a discriminator network is to distinguish the real data from the fake data generated by the generator network. It is in fact a classification issue, and many neural networks have been proposed to perform this task. For example, [Zhang et al. \(2019\)](#) use a 3-layer multilayer perceptron (MLP) as the discriminator in their generative adversarial network, and [Chen et al. \(2024\)](#) use an RNN combined with a FNN. However, both MLP and RNN models suffer from the issue of having a large number of parameters, especially when dealing with high-dimensional data. Therefore, in the selection of the discriminator, we choose to construct it using a combination of a 2-layer CNN and a 3-layer fully connected FNN.

The reason is that CNN is particularly well suited to handle high-dimensional data. The convolutional layers are effective in extracting features and reducing dimensionality. Through the abstraction provided by CNN, the discriminator network can capture critical information with fewer features. Additionally, the parameter sharing characteristic of CNN helps to reduce the number of model parameters and computation time, relieving the training burden. Then, the fully connected FNN integrates the features extracted by the convolutional layers, further processing these features nonlinearly and interactively to map them into a binary classification task (0 or 1). This enables the discriminator network to directly output its judgments on the authenticity of the input data into the loss function. This combination allows the discriminator to distinguish more accurately between real and generated data and improve training efficiency.

In summary, we propose a new GAN model with a generator network consisting of LSTM and self-attention and a discriminator network consisting of CNN and FNN. The primary goal of personalizing GAN is to enhance the model's ability to effectively extract crucial information from high-dimensional datasets and mitigate overfitting issues, so that it achieves superior out-of-sample performance.

3. Description of data and predictors

The data used in our empirical analysis is downloaded from China Stock Market & Accounting Research Database (CSMAR). The sample period of the monthly data spans from January 2003 to December 2022. We replace the missing data by its previous value, and after removing stocks with a duration of less than 5 years, a total of 1543 stocks are included in the following empirical analysis. We use the rolling windows method to generate the training set with a window length of 24 months and the validation set with a validation ratio of 0.1. The data of the last five years are set as a test set. Furthermore, the monthly cross-sectional data are normalized between 0 and 1.

For each stock, we collect 64 firm-level predictors. Meanwhile, in order to analyze the influence of different categorize of predictors on the prediction results, we divide the predictors into seven categories: growth, valuation, profitability, solvency, operations, governance and transaction. [Table 1](#) presents all the predictors, their acronyms and categories.

In addition to firm predictors, ten financial market predictors are included to reflect overall market information. The chosen predictors are partially referenced to [Welch and Goyal \(2008\)](#), with the addition of the A-share turnover rate and the risk-free rate. The specific financial market predictors are listed in [Table 2](#).

4. Empirical analysis

4.1. Prediction performance evaluation

4.1.1. Full sample analysis

We assess the predictive performance of the proposed GAN model by comparing it with five benchmark models: OLS, ENet, NN3, LSTM and another GAN model with a generator of LSTM and a discriminator of RNN based on [Chen et al. \(2024\)](#) (GAN-C for short). The evaluation metrics include out-of-sample R^2 , mean absolute error (MAE), mean squared error (MSE) and root mean squared error (RMSE).

[Table 3](#) shows the in-sample and out-of-sample prediction performances of the six models. The results show that OLS performs the worst, with an out-of-sample R^2 of only -0.0590 . ENet, which considers overfitting issue, shows a slightly better out-of-sample R^2 of -0.0523 compared to OLS. The out-of-sample R^2 for NN3 is 0.3114 and outperforms OLS and ENet because NN3 takes into account the non-linear structures as a deep learning model. LSTM achieves an out-of-sample R^2 of 0.3971 and outperforms OLS

Table 1
Categories, acronyms and description of 64 firm predictors.

Category	Acronym	Description	Category	Acronym	Description
Growth	ca	Capital Accumulation Rate	Solvency	dtar	Days of Turnover of Accounts Receivable
	fag	Fixed Assets Growth Rate		em	Equity Multiplier
	nagps	Net Asset Growth Rate Per Share		etd	Equity to Debt Ratio
	oeg	Owners' Equity Growth Rate		iar	Inventory and Revenue
	opg	Operating Profit Growth Rate		ic	Interest Coverage Ratio
	rg	Revenue Growth Rate		oc	Operating Capital
	sg	Sustainable Growth Rate		qr	Quick Ratio
Valuation	tag	Total Assets Growth Rate	Operation	cat	Current Assets Turnover
	tpg	Total Profit Growth Rate		cati	Current Assets to Income
	bm	Book-To-Market Ratio		cet	Cash and Cash Equivalents Turnover
	mv	Market Value		cint	Capital Intensity
	mvttta	Market Value to Tangible Assets Ratio		dit	Days of Inventory Turnover
	pb	Price-To-Book Ratio		dtap	Days of Turnover of Accounts Payable
	pe	Price Earnings Ratio		fat	Fixed Assets Turnover
Profitability	q	Tobin Q		fati	Fixed Assets to Income
	ctax	Combined Tax Rate	Governance	ncat	Non-Current Assets Turnover
	ebit	Earnings Before Interest and Tax		oct	Operating Capital Turnover
	gom	Gross Operating Margin		tot	Total Assets Turnover
	itax	Income Tax Rate		atcdps	After-Tax Cash Dividend Per Share
	ltroc	Long-term Return on Capital		cdcm	Cash Dividend Cover Multiple
	nom	Net Operating Margin		dm	Dividend Multiple
Solvency	om	Operating Margin		err	Earnings Retention Rate
	ombit	Operating Margin Before Interest and Tax	Transaction	evm	Enterprise Value Multiple
	poa	Profits on Assets		ptcdps	Pre-Tax Cash Dividend Per Share
	roa	Return on Assets		beta	Mariket Beta
	roca	Return on Current Assets		beta_s	Beta Squared
	roe	Return on Equity		lt_reversal	Long-Term Reversal
	rofa	Return on Fixed Assets		mom_12 m	Long-Term Momentum
Solvency	roic	Return on Invested Capital		mom_3 m	Short-Term Momentum
	alr	Asset–Liability Ratio		mom_6 m	Medium-Term Momentum
	cashr	Cash Ratio		st_reversal	Short-Term Reversal
Solvency	cr	Current Ratio		turnover	Turnover Rate

Table 2
Acronyms and description of ten financial market predictors.

Acronym	Description	Acronym	Description
ady	A-Share Dividend Yield	avol	A-Share Volatility
amv	A-share Market Value	bmy_5y	China Bond Maturity Yield :5 Years
apb	A-Share Price-To-Book Ratio	bmy_10	China Bond Maturity Yield :10 Years
ape	A-Share Price Earnings Ratio	mfrer	Monthly Risk-Free Return
aturnover	A-Share Turnover	ts	Term Spread

and ENet significantly in prediction performance. In addition, since it takes into account not only the nonlinear structure but also temporal information, it performs further than NN3. the proposed GAN model further improves the predictive ability of LSTM, achieving an out-of-sample R^2 of 0.5974. Besides, the proposed GAN model still performs best among the six models in terms of the other three metrics. The two models in the GAN framework have the best out-of-sample R^2 performance of 0.5696 and 0.5974, so it can be seen that the GAN framework helps to improve the prediction. While our proposed GAN takes into account the extraction of high-dimensional features and therefore performs slightly better than GAN-C.

Furthermore, we compare the difference between out-of-sample and in-sample R^2 to assess the models' generalization ability. OLS shows a decrease of 0.1284 in R^2 , and ENet decreases by 0.1229. The minor difference in the reduction of R^2 between OLS and ENet indicates that the sparsity contributes little to improving the ability of generalization. NN3 has a decrease of 0.0845 in R^2 and generalizes better than non-machine learning models. LSTM experiences a decrease of 0.0673 in R^2 . The decrease is only about one half of OLS and ENet, showing that the time dependency has an impact on improving the generalization ability. GAN-C further improves generalization on LSTM with an out-of-sample R^2 reduction of 0.0336. The out-of-sample R^2 of the proposed GAN model has only a reduction of 0.0274, which is only 1/3 of that for LSTM. It is worth noting that the proposed GAN model includes both LSTM and self-attention mechanism. Therefore, this result implies that considering both sparsity and time dependence is more likely to prevent overfitting than considering only one of them.

To statistically verify the differences in out-of-sample predictive performance among different models, we utilize the Conditional Superior Predictive Ability (CSPA) test (Li et al., 2022) to conduct pairwise comparisons of predictive performance. Unlike traditional approaches, such as the unconditional predictive ability test (UEPA) proposed by Diebold and Mariano (2002), the CSPA test is more effective in capturing the differences in predictive performance under varying economic conditions. In this analysis, we use the CSI

Table 3

Comparison of the evaluation metrics of OLS, ENet, NN3, LSTM, GAN-C and the proposed GAN model.

Evaluation metrics	In-sample					
	OLS	ENet	NN3	LSTM	GAN-C	Proposed GAN
R^2	0.0694	0.0706	0.3989	0.4644	0.6032	0.6248
$MAE(10^{-2})$	7.1568	7.0602	6.9777	6.6244	5.7860	4.3517
$MSE(10^{-2})$	0.8945	0.8710	0.8347	0.8327	0.6181	0.3489
$RMSE(10^{-2})$	9.4582	9.3331	9.1362	9.1253	7.8618	5.9070
Evaluation metrics	Out-of-sample					
	OLS	ENet	NN3	LSTM	GAN-C	Proposed GAN
R^2	-0.0590	-0.0523	0.3144	0.3971	0.5696	0.5974
$MAE(10^{-2})$	7.2259	7.1037	7.0040	6.9458	5.9194	4.4069
$MSE(10^{-2})$	0.8959	0.8779	0.9401	0.9147	0.6530	0.3653
$RMSE(10^{-2})$	9.4657	9.5700	9.6958	9.5638	8.0805	6.0444

Table 4

Out-of-sample predictive performance test between pairs of models.

	OLS	ENet	NN3	LSTM	GAN-C	Proposed GAN
ENet	-1.2037	–				
NN3	2.9542**	2.8075**	–			
LSTM	3.5016***	3.4063***	0.5589	–		
GAN-C	4.9051***	4.8029***	1.9524*	1.4092	–	
Proposed GAN	5.2048***	5.1083***	2.2086**	1.7041*	0.3057	–

Note: The table reports the CSPA test results between pairs of models; *: $p < 0.05$, **: $p < 0.01$, and ***: $p < 0.001$.**Table 5**Out-of-sample R^2 for each model across different test sets.

	OLS	ENet	NN3	LSTM	GAN-C	Proposed GAN
2015 Stock Crash	-0.1075	-0.0982	0.1785	0.2951	0.4865	0.5123
COVID-19 Pandemic	-0.0850	-0.0755	0.2201	0.3305	0.5112	0.5458

300 index as a proxy for market conditions to evaluate the conditional expected loss difference $h_{12} = E[Y_{12,t}|CSI_{t-1}]$, where $Y_{12,t}$ denotes the loss differential between Model 1 and Model 2.

Table 4 reports the statistical results of the CSPA test, highlighting significant differences in predictive performance among the models. The GAN-based models (GAN-C and the proposed GAN) achieve statistical significance at the 1% level compared to OLS and ENet. The difference between GAN-C and NN3 is significant at the 10% level, whereas the difference between GAN-C and LSTM is not statistically significant. In contrast, the proposed GAN demonstrates significant differences from NN3 and LSTM at the 5% and 10% levels, respectively. These findings suggest that both the proposed GAN and GAN-C substantially outperform traditional models in terms of overall predictive capability, particularly showing a stronger predictive edge over conventional linear methods.

To assess the impact of different test set partitions on our conclusions and ensure the robustness of our results, we conducted additional analyses during two stress periods: the 2015 stock market crash (June to August 2015) and the early phase of the COVID-19 pandemic (January to March 2020). These analyses evaluated the out-of-sample performance of each model. Table 5 presents the out-of-sample R^2 values for each model during these periods. The results indicate that linear models (OLS and ENet) consistently exhibit negative out-of-sample R^2 , suggesting poor predictive capability and an inability to effectively capture the market's complex fluctuations during crises. Although conventional deep learning models (NN3 and LSTM) perform significantly better than linear models, their predictive ability declines considerably during stress periods, revealing limitations in handling extreme market volatility. Due to unforeseen policy interventions during the crash and the pandemic, the generalization performance of GAN-based models (GAN-C and the proposed GAN) is also somewhat affected; however, their intricate nonlinear structures and generative adversarial training mechanisms enable them to outperform other models during crisis periods, demonstrating superior predictive capability. Overall, these results suggest that GAN-based models outperform both traditional linear models and other deep learning models in terms of predictive ability and robustness across varying market conditions.

4.1.2. Subsample analysis

To account for potential heterogeneity in the model's predictive ability and to better capture the unique characteristics of the Chinese stock market, we divided the full sample by ownership type and firm size, and conducted corresponding subsample analyses. The results are summarized in Table 6. Overall, our findings indicate that the proposed GAN model consistently outperforms all benchmark models across the various subsamples, although its performance differs between the two groups within each subsample analysis. The details of the subsample analyses are presented below:

First, we perform a grouped analysis of small-cap stocks (bottom 30% by market value) and large-cap stocks (top 70%). The first two rows of Table 6 present the out-of-sample estimation results, demonstrating the superior performance of the proposed GAN

Table 6
The out-of-sample R^2 of each model under different subsamples..

	OLS	ENet	NN3	LSTM	GAN-C	Proposed GAN
Top 70%	-0.2732	-0.213	0.1167	0.2267	0.3666	0.3971
Bottom 30%	0.0772	0.1066	0.4574	0.4974	0.6767	0.6972
SOE	-0.0819	-0.0524	0.2876	0.3482	0.5181	0.5475
Non-SOE	0.0491	0.0516	0.3892	0.4339	0.6136	0.6342

Table 7
Performances of portfolios derived from OLS, ENet, NN3, LSTM, GAN-C and the proposed GAN model.

	OLS	ENet	NN3	LSTM	GAN-C	Proposed GAN
Long-Short						
<i>Ret</i>	0.0791	0.0789	0.1118	0.1101	0.1225	0.1385
<i>Std</i>	0.0772	0.1150	0.1154	0.0508	0.0532	0.0546
<i>SR</i>	0.7658	0.5123	0.7954	1.7751	1.9257	2.1703
Long-Only						
<i>Ret</i>	0.0335	0.0465	0.0728	0.1013	0.1141	0.1208
<i>Std</i>	0.0903	0.1270	1.1205	0.0571	0.0575	0.0604
<i>SR</i>	0.1493	0.2085	0.4385	1.4237	1.6359	1.6683

model in both small-cap and large-cap samples. Notably, the model's performance is significantly better in the small-cap group, achieving an out-of-sample R^2 of 0.6972, compared to 0.3971 in the large-cap group. These findings further support the conclusions of Leippold et al. (2022) and Allen et al. (2024) regarding the Chinese stock market.

Secondly, considering that state-owned enterprises (SOEs) constitute approximately 41% of the market and differ significantly from non-state-owned enterprises (non-SOEs) in several key aspects, it is crucial to compare the predictive performance of the models between these two groups. For this purpose, we divided the full sample into SOEs and non-SOEs for further analysis, with the last two rows of Table 6 presenting the out-of-sample predictive performance for each model across these groups. Most notably, the proposed GAN model consistently demonstrates the best predictive performance in both the SOE and non-SOE groups. Furthermore, we observe that the models perform better for non-SOEs than for SOEs, which may be attributed to the fact that SOEs, as state-controlled entities, pursue not only profit but also national policy objectives. This dual mandate complicates the predictability of their returns, requiring highly flexible methods to capture their nonlinear dynamics. It is also worth noting that, as SOEs typically dominate key sectors such as banking, infrastructure, and defense, and tend to have larger market capitalizations, the overlap between SOEs and large-cap stocks helps explain the similarities observed in the results from both subsample analyses.

In summary, our findings suggest that predicting monthly returns for small-cap (non-SOE) stocks in the A-share market is generally easier compared to large-cap (SOE) stocks across nearly all models. Additionally, the GAN-based neural network model consistently demonstrates superior predictive performance across the various subsamples.

4.2. Economic performance evaluation

In this subsection, we evaluate the model's performance in economic value. Specifically, we compare the Sharpe ratios for portfolios constructed using the predicted returns generated by different models.

We construct two types of portfolios based on the returns predicted by different models. The first type is a long-short portfolio. For each model, we calculate the cumulative return of each stock over the past three years during the in-sample period, as well as for the first month of the out-of-sample period. We then rank the stocks based on their cumulative returns and divide them into deciles, going long on the stocks in the highest decile (decile 1) and short on those in the lowest decile (decile 10). These portfolios are rebalanced monthly based on market value. The second portfolio type is a long-only portfolio. Due to strict short-selling constraints, implementing long-short strategies is nearly impossible in the Chinese stock market. Therefore, we also consider a long-only scenario, where we hold only the stocks in the highest decile (decile 1), making this approach potentially more practical for market participants in China.

The results in Table 7 indicate that the portfolio constructed using the proposed GAN model delivers the best performance, with annualized returns of 13.85% and 12.08% for the long-short and long-only portfolios, respectively, and Sharpe ratios of 2.1703 and 1.6683. The GAN-C, LSTM, and NN3 models exhibit slightly lower performance compared to the proposed GAN model. In contrast, the traditional machine learning models (OLS and ENet) show a significant performance gap, with annualized returns of only 7.91% and 7.89% for their respective long-short portfolios, and Sharpe ratios of 0.7658 and 0.5123. The performance of the long-only portfolios for OLS and ENet shows an even greater decline. Overall, the long-short strategies outperform the long-only strategies, as the inability to engage in short-selling limits the capacity of long-only portfolios to hedge against unfavorable market movements.

Fig. 3 shows the cumulative returns of the portfolios for each model during the out-of-sample period. The proposed GAN model demonstrates superior performance, achieving cumulative returns of 48.24% for the long-short portfolio and 40.81% for the long-only portfolio, whereas OLS, ENet, and NN3 exhibit negative cumulative returns during many out-of-sample periods. The evaluation of model performance based on portfolio returns and Sharpe ratios is consistent with the out-of-sample R^2 results, further confirming that the proposed GAN model outperforms the other models.

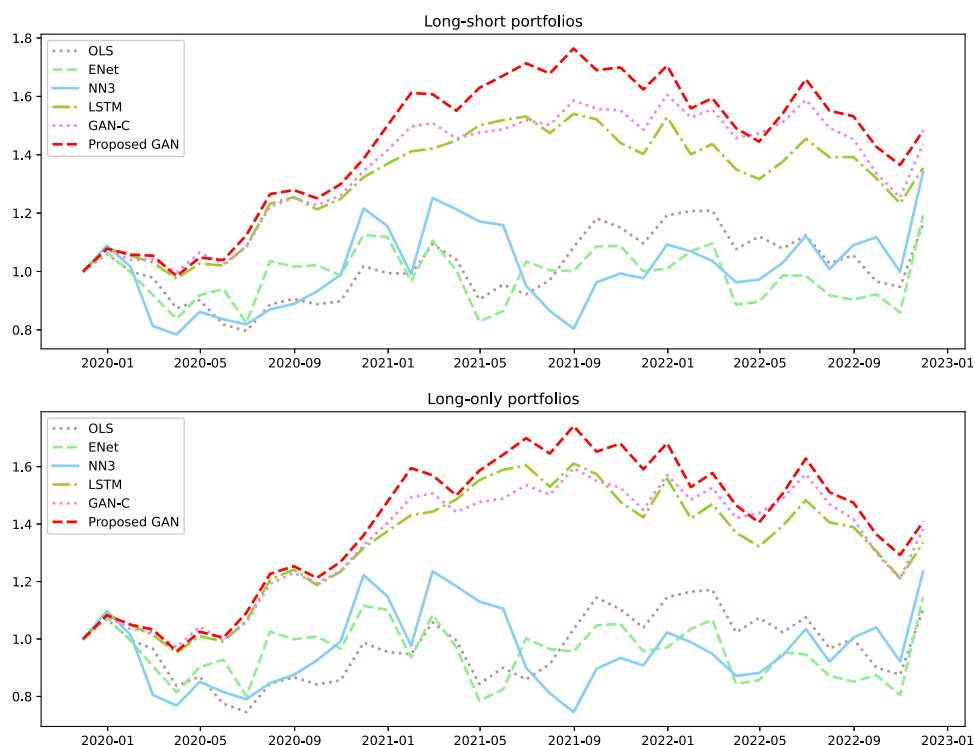


Fig. 3. Out-of-sample cumulative returns of the portfolios derived from OLS, ENet, NN3, LSTM, GAN-C and the proposed GAN model.

Table 8

Regressions of portfolio returns on the FF3 and FF5 models.

	FF3						FF5					
	OLS	ENet	NN3	LSTM	GAN-C	Proposed GAN	OLS	ENet	NN3	LSTM	GAN-C	Proposed GAN
Long-Short												
α	0.0063 (0.4790)	-0.0010 (0.9510)	0.0056 (0.7540)	0.0068 (0.0670)*	0.0104 (0.0120)**	0.0111 (0.0150)**	0.0100 (0.2690)	0.0013 (0.9390)	-0.0032 (0.8570)	0.0067 (0.0720)*	0.0117 (0.0030)***	0.0128 (0.0080)**
β	1.2192 (0.0000)***	1.4616 (0.0000)***	1.7012 (0.0000)***	1.0152 (0.0000)***	0.9514 (0.0000)***	0.9784 (0.0000)***	1.1682 (0.0000)***	1.4596 (0.0000)***	1.8526 (0.0000)***	1.0000 (0.0000)***	0.9086 (0.0000)***	0.9546 (0.0000)***
SMB	-0.5000 (0.0270)**	-0.6169 (0.1190)	-0.5828 (0.1890)	-0.1392 (0.1210)	-0.3332 (0.0020)***	-0.4283 (0.0000)***	-0.8280 (0.0070)***	-0.7603 (0.6520)	0.2566 (0.1620)	-0.1656 (0.1640)	-0.5268 (0.0000)***	-0.5554 (0.0000)***
HML	1.0950 (0.0000)***	1.1688 (0.0200)**	1.5234 (0.0080)***	0.1734 (0.1200)	-0.1952 (0.1280)	0.1745 (0.1640)	0.9269 (0.0210)**	1.3930 (0.0580)*	2.2582 (0.0050)***	-0.0033 (0.9830)	-0.4801 (0.0090)***	0.0508 (0.7780)
RMW	-	-	-	-	-	-	-1.1668 (0.1270)	-1.3138 (0.3490)	2.1642 (0.1510)	0.3464 (0.2610)	-0.1878 (0.5770)	-0.2937 (0.4040)
CMA	-	-	-	-	-	-	-0.3184 (0.6780)	-1.1978 (0.4030)	-0.2681 (0.8590)	0.5545 (0.0820)*	0.4721 (0.1750)	0.0851 (0.8120)
Long-Only												
α	0.0062 (0.5363)	0.0077 (0.6719)	0.012 (0.4722)	0.0058 (0.1570)	0.0095 (0.0313)**	0.0091 (0.0451)**	0.0106 (0.2997)	0.0033 (0.6269)	0.0023 (0.6277)	0.0055 (0.1794)	0.0108 (0.0177)***	0.0108 (0.0152)***
β	1.4233 (0.0000)***	1.6606 (0.0002)***	1.5474 (0.0002)***	1.1302 (0.0000)***	1.0524 (0.0000)***	1.0314 (0.0000)***	1.3629 (0.0000)***	1.6119 (0.0005)***	1.6419 (0.0001)***	1.1129 (0.0000)***	1.0214 (0.0000)***	0.9856 (0.0000)***
SMB	-0.5963 (0.0210)**	-0.4119 (0.3617)	-0.6329 (0.1295)	-0.1827 (0.0761)*	-0.4595 (0.0001)***	-0.3699 (0.0017)***	-0.9812 (0.0052)***	-0.6251 (0.3203)	-0.1938 (0.7286)	-0.2093 (0.1166)	-0.6054 (0.0001)***	-0.5781 (0.0002)***
HML	1.2983 (0.0002)***	1.0154 (0.0769)*	1.4123 (0.0089)***	0.1927 (0.1313)	0.1673 (0.2102)	-0.2415 (0.0841)*	1.0971 (0.0170)**	0.6480 (0.4375)	2.0726 (0.0084)***	-0.0426 (0.8069)	-0.0454 (0.8091)	-0.5706 (0.0039)***
RMW	-	-	-	-	-	-	-1.3574 (0.1204)	-0.0737 (0.9638)	0.4073 (0.7792)	0.4923 (0.1544)	-0.1530 (0.6774)	-0.1506 (0.6760)
CMA	-	-	-	-	-	-	-0.3590 (0.6799)	0.7050 (0.6689)	-1.1118 (0.4523)	0.7567 (0.0345)*	0.3458 (0.3568)	0.5837 (0.1172)

Note: In parentheses, we list the p-values of the regression coefficients; *: $p < 0.05$, **: $p < 0.01$, and ***: $p < 0.001$.

Table 9

Top 20 important firm predictors of OLS, ENet, NN3, LSTM, GAN-C and the proposed GAN model.

Category	Acronym	OLS	Enet	NN3	LSTM	GAN-C	Proposed GAN	Category	Acronym	OLS	Enet	NN3	LSTM	GAN-C	Proposed GAN
Growth	ca							Solvency	dtar		19		20	15	8
	fag								em						
	nagps		16						etd						
	oeg								iar						
	opg			2	9		19		ic			15	19		
	rg	4	5	4					oc						
	sg				14	11	10		qr						
Valuation	tag		18	19				Operation	cat						
	tpg	10			16				cati						
	bm	19	7	10	4	3	2		cet						
	mv								cint						
	mvttta		11						dit						
	pb								dtap						
	pe								fat						
Profitability	q							Governance	fati						
	ctax			7	10	17	18		ncat						
	ebit			1					oct						
	gom	11	6		1	1	1		tot						
	itax	13	9		15				atcdps	18					
	ltroc	12	4		18	12	14		cdcm	15	12	9			
	nom	1	15	8					dm		13				
	om	8		20		20		Transaction	err				7	7	16
	ombit			13					evm						
	poa			16	13	18	15		ptcdps	20	20				
Solvency	roa	5			17	16	13		beta			6	11	8	6
	roca	14							beta_s					19	17
	roe	16		12		10	9		lt_reversal	3	1	3	6	6	7
	rofa	6		17					mom_12 m	9	10	5	3	4	5
	roic	2	17	14		14	20		mom_3 m				8	9	12
	alr		14	18		13	11		mom_6 m	7	2	11	2	2	4
	cashr								st_reversal	17	3		12		
	cr								turnover		8		5	5	3

Note: The numbers 1–20 represent the ranking of predictors importance, where 1 represents the most important predictor, and the importance decreases sequentially.

Next, we assess whether the portfolios constructed using these models generate excess returns that cannot be explained by traditional pricing models, such as the Fama–French multi-factor models. If the returns predicted by machine learning models are viewed as a deep factor, the presence of excess returns would indicate that this deep factor represents an anomaly in the A-share market.

To test for the existence of excess returns, we regress the portfolio returns generated by the six models on the traditional Fama–French three-factor (FF3) and five-factor (FF5) models. The regression results are reported in Table 8. For the long-short portfolios, the α values of the FF3 and FF5 models for OLS, ENet, and NN3 are not statistically significant, suggesting that the returns of these portfolios can be explained by the multi-factor models. In contrast, the α values for LSTM are 0.68% and 0.67% for the FF3 and FF5 models, respectively, and are statistically significant at the 10% level. For GAN-C, the α values are 1.04% and 1.17%, with statistical significance at the 5% and 1% levels, respectively. The proposed GAN model also shows statistical significance at the 5% and 1% levels, with α values increasing to 1.11% and 1.28%, the highest among all models. Similar conclusions apply to the long-only portfolios, with significant excess returns observed; therefore, we do not elaborate further on these results here.

Based on the above results, we conclude that the portfolios constructed using the proposed GAN model can generate significant excess returns that cannot be explained by multi-factor models. In other words, the returns generated by the proposed GAN model represent an anomaly in the Chinese A-share market.

4.3. Predictors importance

In this subsection, we investigate the importance of predictors for prediction accuracy. We follow (Kelly et al., 2019) to define the predictor importance, which is measured as the reduction in the out-of-sample R^2 when the value for a specific predictor is set to zero while keeping the model estimates unchanged.

We first investigate the importance of firm predictors. Fig. 4 presents the importance of all firm predictors under different models. The lighter colors indicate greater reduction in out-of-sample R^2 and higher predictors importance. Fig. 5 shows the importance of different predictor categories. To facilitate comparison, the predictor importance values have been normalized so that the most important predictor has an important value of one.

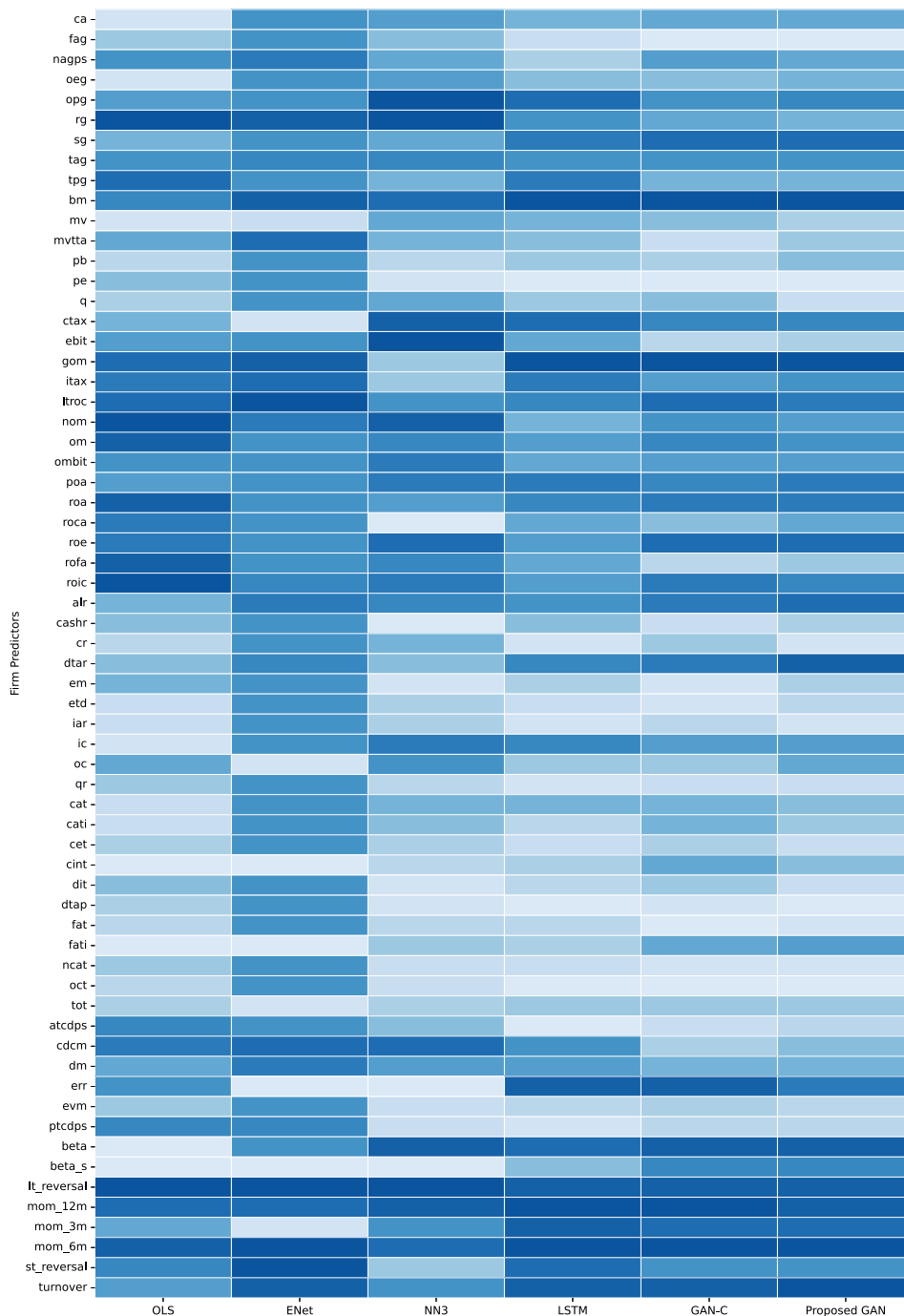


Fig. 4. Importance rankings of all firm predictors.

From the color distribution in Fig. 4 and the results in Fig. 5, it can be observed that the importance of predictor categories is relatively consistent across different models. The two most important categories are transaction and profitability, while the least important categories are solvency and operation. Transaction category plays an important role as it directly provides key information on market sentiment, supply and demand dynamics and liquidity. The result that transaction predictors have the highest contribution implies that investment in the Chinese A-share market should value technical analysis. Besides, the results that profitability predictors are the second important category of predictors imply the importance of value investing in the Chinese stock market.

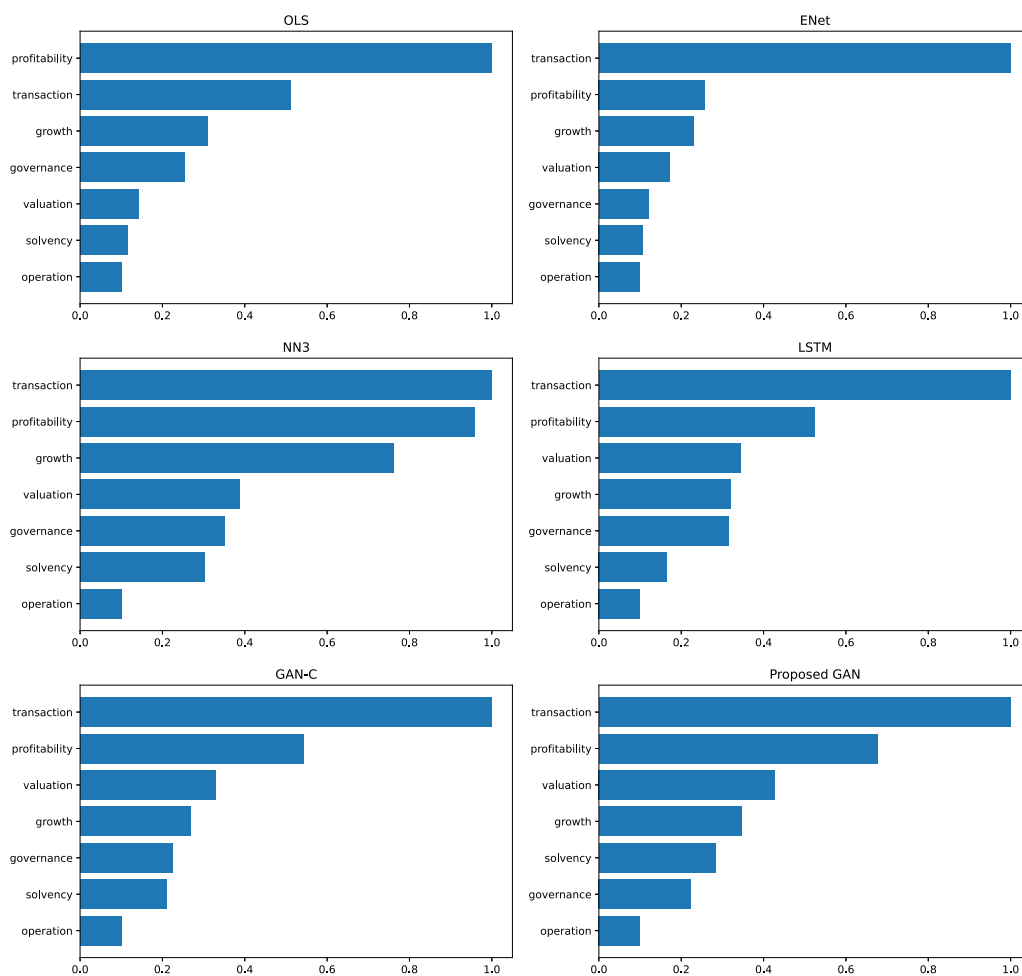


Fig. 5. Category importance of firm predictors for OLS, ENet, NN3, LSTM, GAN-C and the proposed GAN model. The importance is normalized by setting the importance of top ranking to one.

On the other hand, profitability category directly affects investors' expectations and is the most important drivers of the intrinsic value of stocks. Hence, its high importance is in line with expectations.

Furthermore, we label the top 20 important firm predictors in Table 9. We can see that LSTM, GAN-C and the proposed GAN have relative commonality in the top 20 important firm predictors, probably because their underlying models are all LSTM, whereas OLS, ENet and NN3 have relatively distinct important predictors and rankings. While their rankings vary across models, most of the top 20 important firm predictors belong to the same categories for all models. Some important predictors, such as *turnover*, *mom_6m* and *lt_reversal*, which reflect liquidity and momentum, are consistent with the conclusions of Kelly et al. (2019) and Gu et al. (2021). On the other hand, some valuation predictors, such as *ep*, that are important in Kelly et al. (2019) and Gu et al. (2021), do not make the top 20 important predictors in the Chinese A-share market, and some probability predictors, such as *roe*, that do not appear in the list of top 20 predictors in Kelly et al. (2019) and Gu et al. (2021), becomes importance in Chinese A-share market. As a result, from a long-term perspective, investors in the Chinese A-share market are more concerned about profitability predictors than valuation predictors.

Fig. 6 displays the importance rankings for financial market predictors. The rankings are generally consistent among the five models. Dividend yield and market value are the most two important predictors in the Chinese A-share market. When overall market value is high, investors tend to be confident in the market profitability and the market is liquid. This helps to drive up stock returns. A higher dividend yield is more attractive to investors and shows stronger profitability and financial stability, further enhancing investor confidence in the stock market.

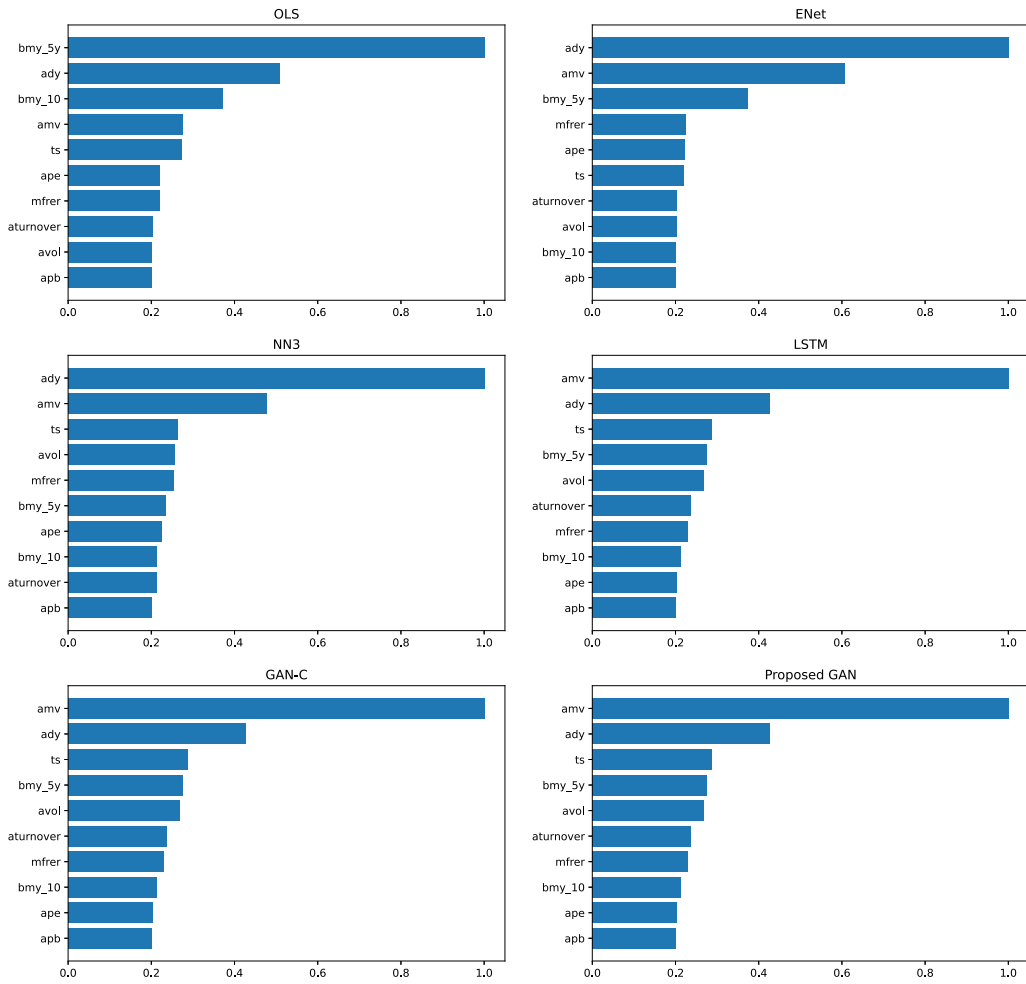


Fig. 6. Importance of financial market predictors for OLS, ENet, NN3, LSTM, GAN-C and the proposed GAN model. The importance is normalized by setting the importance of top ranking to one.

4.4. Macroeconomic mechanism analysis

The results in Section 4.2 show that the return of the long-short investment portfolios constructed using the proposed GAN model (referred to as the deep factor) is an anomaly factor in the Chinese A-share market. However, Avramov et al. (2023) show that the payoff of deep learning models often becomes statistically insignificant in various market states. Thus, we also investigate whether our deep factor is influenced by different macroeconomic states. This is an important aspect in assessing the robustness of the application of the deep factor. As for variable selection, based on previous study (Huang et al., 2015; Jiang et al., 2018) and actual situation of the Chinese market, we introduce a richer set of macroeconomic variables related to investment, consumption, exports, inflation and policies, to reflect different macroeconomic states in our study. The specific variables are new fixed assets (*nfa*), social financing scale (*sfs*), total retail sales (*trs*), consumer satisfaction index (*csi*), foreign trade throughput (*ftt*), economic policy uncertainty index (*epui*), consumer price index (*cpi*) and M2 growth rate (*m2*). The following regression model is used to evaluate the impact of changes in the macroeconomic state on the deep factor:

$$DF_t = \alpha + \gamma Macro_Dummy_{t-1} + \beta_{MKT} MKT_{t-1} + \beta_{SMB} SMB_{t-1} + \beta_{HML} HML_{t-1} + \beta_{RMW} RMW_{t-1} + \beta_{CMA} CMA_{t-1} + \varepsilon_t \quad (6)$$

where DF_t represents the deep factor generated by the proposed GAN model, $Macro_Dummy_{t-1}$ is a specific macroeconomic variable defined as a dummy variable that takes a value of one if its value is above the median over the sample period and zero otherwise. The other five variables (MKT , SMB , HML , RMW and CMA) from FF5 model are control variables in the regression.

Table 10 presents the results of the regression. From the perspective of investment-related variables, when new fixed assets (*nfa*) are in the high state (i.e., taking a value of one), the deep factor leads to an increase in excess returns of 2.17%, which is statistically significant at the 10% level. Generally, an increased investment in fixed assets is considered as a signal of economic

Table 10

Impact of different macroeconomic states on a deep factor generated by the proposed GAN model.

	<i>nfa</i>	<i>sfs</i>	<i>trs</i>	<i>csi</i>	<i>ftt</i>	<i>epu</i>	<i>cpi</i>	<i>m2</i>
α	0.0026 (0.7420)	0.0250 (0.0070)***	0.0138 (0.0000)***	0.0029 (0.4720)	0.0036 (0.6660)	0.0143 (0.0000)***	0.0139 (0.0000)***	0.0077 (0.3150)
γ	0.0217 (0.0850)*	-0.0250 (0.0420)**	-0.0093 (0.0940)*	0.0119 (0.0250)**	0.0213 (0.0840)*	-0.0100 (0.0600)**	-0.0085 (0.1220)	0.0160 (0.2180)
β_{MKT}	-0.1268 (0.4110)	-0.0799 (0.5980)	0.9594 (0.0000)***	0.9284 (0.0000)***	-0.0849 (0.5810)	0.9442 (0.0000)***	0.9443 (0.0000)***	-0.0795 (0.6090)
β_{SMB}	-0.3559 (0.1550)	-0.2239 (0.3560)	-0.4132 (0.0000)***	-0.4101 (0.0000)***	0.0062 (0.9800)	-0.4285 (0.0000)***	-0.4737 (0.0000)***	0.0193 (0.9390)
β_{HML}	-0.0758 (0.8000)	0.0081 (0.9780)	0.0414 (0.7470)	0.0537 (0.6660)	-0.1622 (0.5910)	0.0317 (0.8040)	0.0812 (0.5210)	-0.1837 (0.5480)
β_{RMW}	-0.1130 (0.7910)	0.0035 (0.9930)	-0.2701 (0.1510)	-0.2747 (0.1340)	-0.0691 (0.8740)	-0.2752 (0.1390)	-0.3823 (0.0430)**	-0.0046 (0.9920)
β_{CMA}	-0.1361 (0.7710)	-0.1179 (0.7990)	-0.1488 (0.4580)	-0.1239 (0.5310)	0.0231 (0.9610)	-0.1145 (0.5690)	-0.2235 (0.2720)	0.0863 (0.8570)

Note: In parentheses, we list the p-values of the regression coefficients; *: $p < 0.05$, **: $p < 0.01$, and ***: $p < 0.001$.**Table 11**

Performances of the proposed GAN model trained with different combinations of predictors.

Predictors combinations	Out-of-sample R^2	Dummy variables							
		<i>nfa</i>	<i>sfs</i>	<i>trs</i>	<i>csi</i>	<i>ftt</i>	<i>epu</i>	<i>cpi</i>	<i>m2</i>
$fc \otimes mkt$	0.5974	0.0217 (0.0850)*	-0.0250 (0.0420)**	-0.0093 (0.0940)*	0.0119 (0.0250)**	0.0213 (0.0840)*	-0.0100 (0.0600)**	-0.0085 (0.1220)	0.0160 (0.2180)
$fc \otimes marco$	0.5711	0.0029 (0.6550)	-0.0024 (0.7200)	-0.0013 (0.8510)	0.0079 (0.2140)	-0.0063 (0.3290)	-0.0007 (0.9280)	-0.0010 (0.8830)	0.0295 (0.1390)
$fc \otimes mkt \otimes marco$	0.6269	-0.0007 (0.9680)	-0.0059 (0.8520)	0.0162 (0.6200)	0.0067 (0.8290)	0.0036 (0.8260)	-0.0060 (0.7630)	-0.0021 (0.9080)	0.0154 (0.3580)

Note: In parentheses, we list the p-values of the regression coefficients; *: $p < 0.05$, **: $p < 0.01$, and ***: $p < 0.001$; $fc \otimes mkt$ presents GAN model trained with firm and financial market predictors. $fc \otimes marco$ and $fc \otimes mkt \otimes marco$ have similar meaning.

recovery, leading to active participation in the stock market. When social financing scale (*sfs*) is in the high state, it decreases the deep factor by 2.50% on average, which is statistically significant at the 5% level. This may be because the central bank may adopt a tightening monetary policy to control excessive credit expansion or curb inflation risks. At the same time, an increase in social financing can divert funds from the stock market to other sectors. Turning to consumption variables, when total retail sales of consumer goods (*trs*) are above the median, there is a decrease in excess returns of 0.93%. Typically, an increase in total retail sales indicates market prosperity, but it can also imply rising inflationary pressures. If the market expects the central bank to adopt a tightening monetary policy, it can have a negative impact on the stock market, while the consumer satisfaction index (*csi*), directly reflecting the willingness to consume, has a significant positive impact on the deep factor. Foreign trade throughput (*ftt*) has a significant positive effect on the deep factor, increasing by 2.13% ($p = 0.0840$), while the excess return of the deep factor decreases by 1.00% ($p = 0.0600$) when economic policy uncertainty (*epu*) increases.

The results in Table 10 indicate that the deep factor constructed by the proposed GAN model can be influenced by macroeconomic changes. It is natural to consider incorporating these macroeconomic variables into the training of the GAN model to further improve the robustness of the deep factor. Therefore, we combine firm, financial market and macroeconomic predictors, and retrain the proposed GAN model under different predictor combinations. Table 11 presents the training and regression results.

It can be observed that the model trained with all three sets of predictors achieves the highest out-of-sample R^2 . This observation confirms that incorporating additional macroeconomic information into firm and financial market information indeed improves the prediction accuracy of the proposed GAN model. It should be noted that once macroeconomic predictors are included in the training, the coefficients of macroeconomic dummy variables in the regression are no longer significant. Furthermore, comparing with $fc \otimes mkt$, almost all p values for γ in $fc \otimes mkt \otimes marco$ increase when the macroeconomic predictors are included in the training.

In all, we conclude that the proposed GAN model including the macroeconomic predictors can effectively capture the macroeconomic information, and the performance of the long-short portfolio constructed from the new GAN model is robust to macroeconomic fluctuations.

5. Conclusion

In this paper, we propose a new GAN model that combines LSTM, self-attention mechanism, CNN and FNN for stock return prediction, and compare the out-of-sample performance of this proposed GAN model with five other benchmark models. By leveraging extensive financial data covering firm, financial market and macroeconomic predictors in the Chinese A-share market, we construct a robust set of predictors for out-of-sample stock return prediction. the proposed GAN model, as an application of complex machine learning techniques, shows better performance in terms of prediction accuracy and portfolio Sharpe ratios.

Our empirical results show that transaction predictors contribute most to the stock return prediction in the Chinese A-share market, suggesting the inefficiency of Chinese stock market and the importance of short-term investment behaviors and technical analysis in return prediction. And the significant contribution of profitability predictors also indicate the importance of value investing in China. Furthermore, the deep factor generated by the proposed GAN model is influenced by macroeconomic states to some extent. After we incorporate more macroeconomic information into model training, these influences are mitigated.

In summary, we show that the proposed GAN model has a good out-of-sample prediction performance and provide new insights into understanding the complexity of the Chinese A-share market. Meanwhile, the deep factor derived from the proposed GAN model provides intuitive and robust guidance for the Chinese A-share investors when making stock investment decisions.

CRediT authorship contribution statement

Hongxu Wu: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Qiao Wang:** Writing – original draft, Investigation, Formal analysis. **Jianping Li:** Writing – review & editing, Supervision, Resources. **Zhibin Deng:** Writing – review & editing, Validation, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China [Grant No. T2293774], China Postdoctoral Science Foundation [Grant No. 2022M713102], the Fundamental Research Funds for the Central Universities [Grant No. E2E49801], and the grant from MOE Social Science Laboratory of Digital Economic Forecast and Policy Simulation at UCAS.

Data availability

Data will be made available on request.

References

- Allen, F., Qian, J.Q., Shan, C., Zhu, J.L., 2024. Dissecting the long-term performance of the Chinese stock market. *J. Finance* 79 (2), 993–1054. <http://dx.doi.org/10.1111/jofi.13312>.
- Avramov, D., Cheng, S., Metzker, L., 2023. Machine learning vs. Economic restrictions: Evidence from stock return predictability. *Manage. Sci.* 69 (5), 2587–2619. <http://dx.doi.org/10.1287/mnsc.2022.4449>.
- Bybee, L., Kelly, B.T., Su, Y., 2022. Narrative Asset Pricing: Interpretable Systematic Risk Factors from News Text. (3895277), Rochester, NY, <http://dx.doi.org/10.2139/ssrn.3895277>.
- Carhart, M.M., 1997. On persistence in mutual fund performance. *J. Finance* 52 (1), 57–82. <http://dx.doi.org/10.1111/j.1540-6261.1997.tb03808.x>.
- Chen, L., Pelger, M., Zhu, J., 2024. *Manage. Sci.* 70 (2), 714–750.
- Daniel, K., Hirshleifer, D., Sun, L., 2020. Short- and long-horizon behavioral factors. *Rev. Financ. Stud.* 33 (4), 1673–1736. <http://dx.doi.org/10.1093/rfs/hhz069>.
- Diebold, F.X., Mariano, R.S., 2002. Comparing predictive accuracy. *J. Bus. Econom. Statist.* 20 (1), 134–144. <http://dx.doi.org/10.1198/073500102753410444>.
- Fama, E.F., French, K.R., 1993. Common risk factors in the returns on stocks and bonds. *J. Financ. Econ.* 33 (1), 3–56. [http://dx.doi.org/10.1016/0304-405X\(93\)90023-5](http://dx.doi.org/10.1016/0304-405X(93)90023-5).
- Fama, E.F., French, K.R., 2015. A five-factor asset pricing model. *J. Financ. Econ.* 116 (1), 1–22. <http://dx.doi.org/10.1016/j.jfineco.2014.10.010>.
- Feng, G., He, J., Polson, N.G., Xu, J., 2024. Deep learning in characteristics-sorted factor models. *J. Financ. Quart. Anal.* 59 (7), 3001–3036.
- Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., 2014. Generative adversarial nets. *Adv. Neural Inf. Process. Syst.* 27.
- Green, J., Hand, J.R.M., Zhang, X.F., 2013. The supraview of return predictive signals. *Rev. Account. Stud.* 18 (3), 692–730. <http://dx.doi.org/10.1007/s11142-013-9231-1>.
- Gu, S., Kelly, B., Xiu, D., 2020. Empirical asset pricing via machine learning. In: Karolyi, A. (Ed.), *Rev. Financ. Stud.* 33 (5), 2223–2273. <http://dx.doi.org/10.1093/rfs/hhaa009>.
- Gu, S., Kelly, B., Xiu, D., 2021. Autoencoder asset pricing models. *J. Econometrics* 222 (1), 429–450. <http://dx.doi.org/10.1016/j.jeconom.2020.07.009>.
- Hou, K., Xue, C., Zhang, L., 2015. Digesting anomalies: An investment approach. *Rev. Financ. Stud.* 28 (3), 650–705. <http://dx.doi.org/10.1093/rfs/hhu068>.
- Huang, D., Jiang, F., Tu, J., Zhou, G., 2015. Investor sentiment aligned: A powerful predictor of stock returns. *Rev. Financ. Stud.* 28 (3), 791–837. <http://dx.doi.org/10.1093/rfs/hhu080>.
- Jiang, F., Tang, G., Zhou, G., 2018. Firm characteristics and Chinese stocks. *J. Manag. Sci. Eng.* 3 (4), 259–283. <http://dx.doi.org/10.3724/SP.J.1383.304014>.
- Kelly, B.T., Pruitt, S., Su, Y., 2019. Characteristics are covariances: A unified model of risk and return. *J. Financ. Econ.* 134 (3), 501–524. <http://dx.doi.org/10.1016/j.jfineco.2019.05.001>.
- Kozak, S., Nagel, S., Santosh, S., 2020. Shrinking the cross-section. *J. Financ. Econ.* 135 (2), 271–292. <http://dx.doi.org/10.1016/j.jfineco.2019.06.008>.
- Kumar, A., Alsadoon, A., Prasad, P.W.C., Abdullah, S., Rashid, T.A., Pham, D.T.H., Nguyen, T.Q.V., 2022. Generative adversarial network (GAN) and enhanced root mean square error (ERMSE): Deep learning for stock price movement prediction. *Multimedia Tools Appl.* 81 (3), 3995–4013. <http://dx.doi.org/10.1007/s11042-021-11670-w>.
- Leippold, M., Wang, Q., Zhou, W., 2022. Machine learning in the Chinese stock market. *J. Financ. Econ.* 145 (2), 64–82. <http://dx.doi.org/10.1016/j.jfineco.2021.08.017>.
- Lettau, M., Pelger, M., 2020. Factors that fit the time series and cross-section of stock returns. In: Van Nieuwerburgh, S. (Ed.), *Rev. Financ. Stud.* 33 (5), 2274–2325. <http://dx.doi.org/10.1093/rfs/hhaa020>.

- Li, J., Liao, Z., Quaedvlieg, R., 2022. Conditional superior predictive ability. In: Molinari, F. (Ed.), *Rev. Econ. Stud.* 89 (2), 843–875. <http://dx.doi.org/10.1093/restud/rdab039>.
- Li, W., Mei, F., 2020. Asset returns in deep learning methods: An empirical analysis on SSE 50 and CSI 300. *Res. Int. Bus. Finance* 54, 101291. <http://dx.doi.org/10.1016/j.ribaf.2020.101291>.
- Ma, T., Leong, W.J., Jiang, F., 2023. A latent factor model for the Chinese stock market. *Int. Rev. Financ. Anal.* 87, 102555. <http://dx.doi.org/10.1016/j.irfa.2023.102555>.
- Novy-Marx, R., 2013. The other side of value: The gross profitability premium. *J. Financ. Econ.* 108 (1), 1–28. <http://dx.doi.org/10.1016/j.jfineco.2013.01.003>.
- Pohl, W., Schmedders, K., Wilms, O., 2018. Higher order effects in asset pricing models with long-run risks. *J. Finance* 73 (3), 1061–1111. <http://dx.doi.org/10.1111/jofi.12615>.
- Principe, T.-V., 2021. Deep reinforcement learning in quantitative algorithmic trading: A review. [arXiv:2106.00123](https://arxiv.org/abs/2106.00123).
- Stambaugh, R.F., Yuan, Y., 2017. Mispricing factors. *Rev. Financ. Stud.* 30 (4), 1270–1315. <http://dx.doi.org/10.1093/rfs/hhw107>.
- Welch, I., Goyal, A., 2008. A comprehensive look at the empirical performance of equity premium prediction. *Rev. Financ. Stud.* 21 (4), 1455–1508. <http://dx.doi.org/10.1093/rfs/hhm014>.
- Zhang, K., Zhong, G., Dong, J., Wang, S., Wang, Y., 2019. Stock market prediction based on generative adversarial network. 2018 International Conference on Identification, Information and Knowledge in the Internet of Things, *Procedia Comput. Sci.* 2018 International Conference on Identification, Information and Knowledge in the Internet of Things, 147, 400–406. <http://dx.doi.org/10.1016/j.procs.2019.01.256>,